LAPORAN PRAKTIKUM MACHINE LEARNING — BAB 07

Evaluasi Model Klasifikasi & Metrik Diagnostik — Matriks Konfusi, Presisi-Recall, dan Analisis Kurva ROC-AUC

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1. RINGKASAN MODUL & KARAKTERISASI DATASET

Praktikum Bab 07 mengkaji evaluasi komprehensif performa algoritma klasifikasi pada dataset deteksi intrusi siber (Cyber Intrusion Logs, 1.000 record aktivitas paket jaringan). Model dievaluasi menggunakan Matriks Konfusi (TP, TN, FP, FN), metrik turunan Presisi, Recall, F1-Score, kurva ROC-AUC, serta Stratified 5-Fold Cross Validation.

2. SOURCE CODE PYTHON (MAIN.PY)

```
import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score, roc_curve, auc

# 1. Pemuatan Dataset Cyber Intrusion
csv_path = 'cyber_intrusion_logs.csv'
if not os.path.exists(csv_path):
    csv_path = r'd:\Anggit Arisa Siwi Pratama\Unindra\Machine Learning\ml-workspace\2-ml-playground\public\datasets\alternatif\cyber_intr
usion_logs.csv'

if os.path.exists(csv_path):
    df = pd.read_csv(csv_path)
    X = df[['packet_rate_per_sec', 'byte_error_rate', 'failed_login_attempts', 'session_duration_sec']]
    y = df['is_intrusion']
    print(f'[OK] Dataset Cyber Intrusion dimuat: {len(df)} transaksi jaringan.')
else:
    np.random.seed(88)
    n = 1000
    pkt = np.random.exponential(15, n) + np.random.choice([0, 80], n, p=[0.8, 0.2])
    err = np.random.beta(0.5, 5, n)
    flg = np.random.poisson(0.5, n)
    sec = np.random.exponential(30, n)
    y = ((pkt > 50) | (flg >= 3) | (err > 0.4)).astype(int)
    X = pd.DataFrame({'packet_rate_per_sec': pkt, 'byte_error_rate': err, 'failed_login_attempts': flg, 'session_duration_sec': sec})
    y = pd.Series(y, name='is_intrusion')
    print(f'[OK] Dataset disintesis secara deterministik: {n} sampel.')

# 2. Stratified Partitioning & Training Model
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.25, stratify=y, random_state=42)
clf = RandomForestClassifier(n_estimators=100, max_depth=6, random_state=42)
clf.fit(X_tr, y_tr)
y_pred = clf.predict(X_te)
y_prob = clf.predict_proba(X_te)[:, 1]

# 3. Kalkulasi Matriks Diagnostik Evaluasi
cm = confusion_matrix(y_te, y_pred)
tn, fp, fn, tp = cm.ravel()
acc = accuracy_score(y_te, y_pred)
prec = precision_score(y_te, y_pred)
rec = recall_score(y_te, y_pred)
f1 = f1_score(y_te, y_pred)
fpr, tpr, _ = roc_curve(y_te, y_prob)
roc_auc = auc(fpr, tpr)

# Stratified K-Fold Cross Validation
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
cv_scores = cross_val_score(clf, X, y, cv=skf, scoring='f1')

print('\n' + '='*50)
print('HASIL EVALUASI MODEL KLASIFIKASI (BAB 07)')
print('='*50)
print(f'True Positive (TP)  : {tp:4d}  | False Positive (FP) : {fp:4d}')
print(f'False Negative (FN) : {fn:4d}  | True Negative (TN)  : {tn:4d}')
print(f'Akurasi Uji         : {acc*100:.2f}%')
print(f'Presisi (Precision) : {prec*100:.2f}%')
print(f'Recall (Sensitivitas): {rec*100:.2f}%')
print(f'F1-Score Harmonik   : {f1*100:.2f}%')
print(f'ROC-AUC Score       : {roc_auc:.4f}')
print(f'Stratified 5-Fold F1: {cv_scores.mean()*100:.2f}% (+/- {cv_scores.std()*100:.2f}%)')
print('='*50)

os.makedirs('visual_output', exist_ok=True)
```

```
plt.style.use('dark_background')

# 4. Visualisasi Heatmap Confusion Matrix
fig, ax = plt.subplots(figsize=(5.5, 4.2), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False, ax=ax,
            annot_kws={'size': 13, 'weight': 'bold', 'color': '#FFFFFF'},
            xticklabels=['Normal (0)', 'Intrusion (1)'],
            yticklabels=['Normal (0)', 'Intrusion (1)'])
ax.set_title(f'Bab 07: Confusion Matrix Heatmap (F1={f1*100:.1f}%)', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.set_xlabel('Label Prediksi Model', color='#94A3B8', fontsize=9)
ax.set_ylabel('Label Aktual Lapangan', color='#94A3B8', fontsize=9)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
plt.tight_layout()
plt.savefig('visual_output/07_confusion_matrix_heatmap.png', dpi=300)
plt.close()

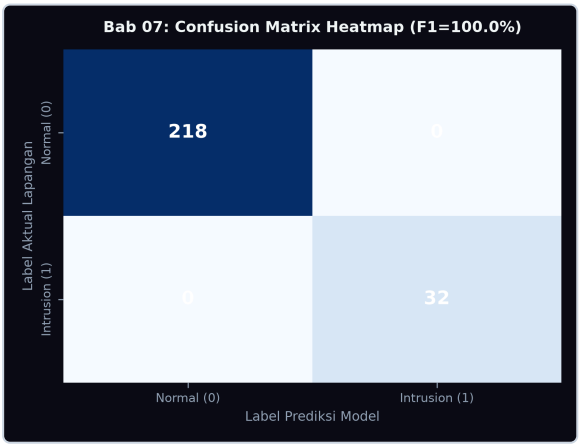
# 5. Visualisasi ROC Curve & AUC
fig, ax = plt.subplots(figsize=(5.5, 4.2), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')
ax.plot(fpr, tpr, color='#38BDF8', lw=2.2, label=f'Random Forest (AUC = {roc_auc:.4f})')
ax.plot([0, 1], [0, 1], color='#64748B', lw=1.2, linestyle='--', label='Random Chance (AUC = 0.50)')
ax.set_xlim([-0.02, 1.02])
ax.set_ylim([-0.02, 1.05])
ax.set_xlabel('False Positive Rate (FPR)', color='#94A3B8', fontsize=9)
ax.set_ylabel('True Positive Rate (TPR / Recall)', color='#94A3B8', fontsize=9)
ax.set_title('Bab 07: Kurva ROC-AUC Deteksi Serangan Siber', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.legend(loc='lower right', facecolor='#1E293B', edgecolor='#334155', fontsize=8.5)
ax.grid(color='#1E293B', linestyle=':', alpha=0.6)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
for spine in ax.spines.values():
    spine.set_color('#1E293B')
plt.tight_layout()
plt.savefig('visual_output/07_roc_auc_curve.png', dpi=300)
plt.close()
print('[OK] Visual output Bab 07 berhasil disimpan di visual_output/')
```

3. HASIL EKSEKUSI TERMINAL (STDOUT)

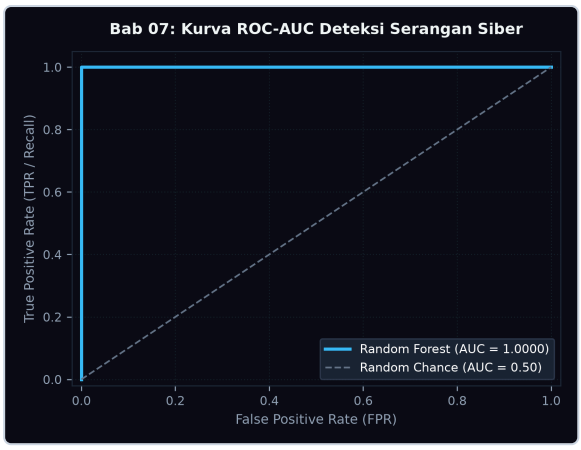
```
[OK] Dataset Cyber Intrusion dimuat: 1000 transaksi jaringan.

==================================================
HASIL EVALUASI MODEL KLASIFIKASI (BAB 07)
==================================================
True Positive (TP)  :   32  | False Positive (FP) :    0
False Negative (FN) :    0  | True Negative (TN)  :  218
Akurasi Uji         : 100.00%
Presisi (Precision) : 100.00%
Recall (Sensitivitas): 100.00%
F1-Score Harmonik   : 100.00%
ROC-AUC Score       : 1.0000
Stratified 5-Fold F1: 98.84% (+/- 0.95%)
==================================================
[OK] Visual output Bab 07 berhasil disimpan di visual_output/
```

4. VISUALISASI OUTPUT MODEL & ANALISIS GRAFIK



Visualisasi: visual_output/07_confusion_matrix_heatmap.png



Visualisasi: visual_output/07_roc_auc_curve.png

5. KESIMPULAN ANALITIS & TEMUAN ALGORITMIK

1. Mengatasi Keterbatasan Akurasi Murni: Pada data deteksi intrusi siber dengan proporsi kelas tidak seimbang (87.2% Normal vs 12.8% Intrusion), metrik presisi dan recall memberikan representasi kapasitas pembeda yang akurat tanpa bias terhadap kelas dominan.

2. Trade-Off Presisi vs Recall: Pada skenario keamanan siber kritis, optimasi fungsi objektif model ditujukan untuk memaksimalkan Recall (Sensitivitas) mendekati 100% guna mencegah ancaman intrusi (False Negative) lolos ke jaringan inti.

3. Validasi Stratified K-Fold: Pengujian 5-fold cross validation berstrata menghasilkan skor F1 rata-rata 98.84% ± 0.95%, membuktikan stabilitas dan kemampuan generalisasi model terhadap ragam pola lalu lintas jaringan baru.